

K. R. JANAHAN

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PROFESSIONAL SUMMARY

Biochemistry undergraduate with hands-on technical proficiency in computational biology, Linux-based environment workflows, genomics, transcriptomics, and structural bioinformatics. Experienced in developing custom **Bash** and **Python** automation scripts for NGS workflows, de novo assemblies, variant identification, and protein-ligand docking. Competent in analyzing high-throughput biological datasets and executing scalable bioinformatics analysis from raw sequence reads to structural visualization.

TECHNICAL SKILLS & COMPETENCIES

Programming & Automation	Python (Biopython, Pandas, NumPy), Bash/Shell Scripting, Linux Command Line (Ubuntu), Git/GitHub Workflows
Genomics & NGS Analysis	Quality Control (FastQC), Read Trimming (fastp, Trimmomatic), Sequence Alignment (BWA, HISAT2), Variant Calling & Annotation (GATK, SAMtools, BCFtools, SnpEff)
Assembly & Annotation	De Novo Bacterial Genome Assembly (SPAdes), Transcriptome Assembly (Trinity), Genome Annotation (Prokka)
Transcriptomics & Expression	RNA-Seq Workflow Execution, Gene Expression Quantification (featureCounts), Differential Expression Pipelines (DESeq2)
Structural Biology & Docking	Macromolecular Structure Retrieval (RCSB PDB), Ligand/Receptor Preparation (Open Babel), Docking Simulations (AutoDock Vina), 3D Visualization (PyMOL, UCSF ChimeraX)
Databases & Utilities	NCBI (BLAST, SRA, Entrez), UniProt, PubMed Literature Mining, IGV (Integrative Genomics Viewer)

INTERNSHIP EXPERIENCE

Bioinformatics Intern | Insilicomics Research Pvt. Ltd.

Ooty, The Nilgiris, Tamil Nadu

- Engineered automated **Bash** scripts for quality assessment, adapter trimming, and data filtering of high-throughput Illumina sequencing reads.
- Implemented reference-guided genome alignments, binary alignment map manipulation, and variant identification pipelines using **BWA** and **SAMtools**.
- Executed transcriptomic processing pipelines, constructing de novo transcript assemblies using **Trinity** for downstream analysis.
- Formulated molecular docking protocols to model ligand-receptor binding interactions and analyze structural binding sites using **AutoDock Vina** and **PyMOL**.
- Developed **Python** utility scripts for parsing multi-FASTA files, processing biological datasets, and streamlining repetitive data extraction tasks.
- Managed project repositories, documentation, and version tracking using **Git** and **GitHub** best practices.

AREAS OF CORE EXPERTISE & TECHNICAL PROJECTS

- Automated WGS & Variant Analysis:** Configured end-to-end Linux/Bash workflows for whole-genome alignment, coverage estimation, SNP/Indel identification, and variant quality score recalibration.
- Transcriptome Assembly & Differential Expression Pipelines:** Built complete transcriptomic analysis workflows covering raw read preprocessing, transcript reconstruction with Trinity, and differential expression profiling.
- De Novo Microbial Genome Assembly & Structural Annotation:** Processed short-read datasets to reconstruct microbial genomes via SPAdes and automatically extract functional gene features and coding sequences via Prokka.
- Protein-Ligand Docking & Virtual Screening Protocols:** Handled small-molecule ligand library preparation, target protein grid box selection, automated batch docking using AutoDock Vina, and interaction interface analysis using PyMOL.
- Bio-Python Utility Scripting & File Automation:** Authored clean, modular Python scripts to parse FASTA/FASTQ headers, extract specific genomic sequences, calculate sequence metrics, and reformat bio-files.

EDUCATION

Bachelor of Science (B.Sc.) in Biochemistry, Genetics, and Biotechnology

Jain University, Bangalore, Karnataka | 2025 – 2028

- Relevant Coursework:** Bioinformatics, Computational Biology, Genomics, Molecular Biology, Structural Biology, NGS Data Analysis, Programming for Biology.

CERTIFICATIONS & CREDENTIALS

Completed Professional Certifications: Documented credentials covering Bioinformatics, Linux/Bash automation, Python script generation, NGS processing, and Molecular Docking workflows

LANGUAGES

English (Professional) | Tamil (Native)